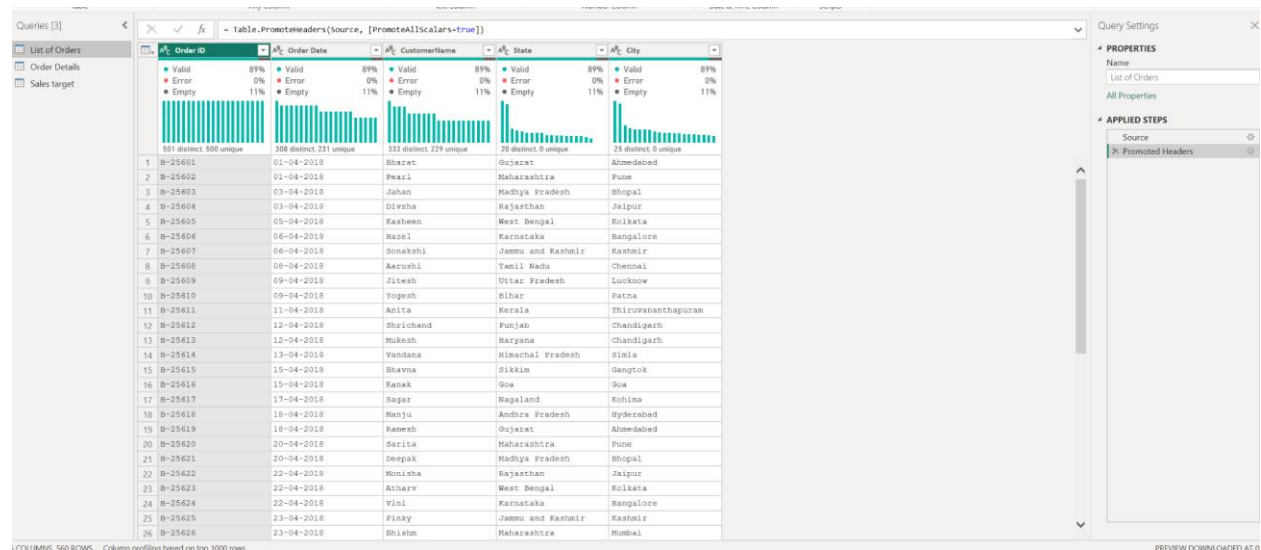


Power BI Assignment 1 – Data Transformation & Data Modeling

Data Import & Transformation

1. Data Import:

- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.



1. Power BI Desktop → Home → Get Data → Text/CSV → select List of Orders.csv → click Transform Data (this opens Power Query Editor instead of loading directly).
2. Inside Power Query Editor: Home → New Source → Text/CSV → Import Order Details.csv.
3. Repeat for Sales Target.csv.

You should now see three queries listed in the left-hand Queries pane: List of Orders, Order Details, Sales Target.

2. Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.

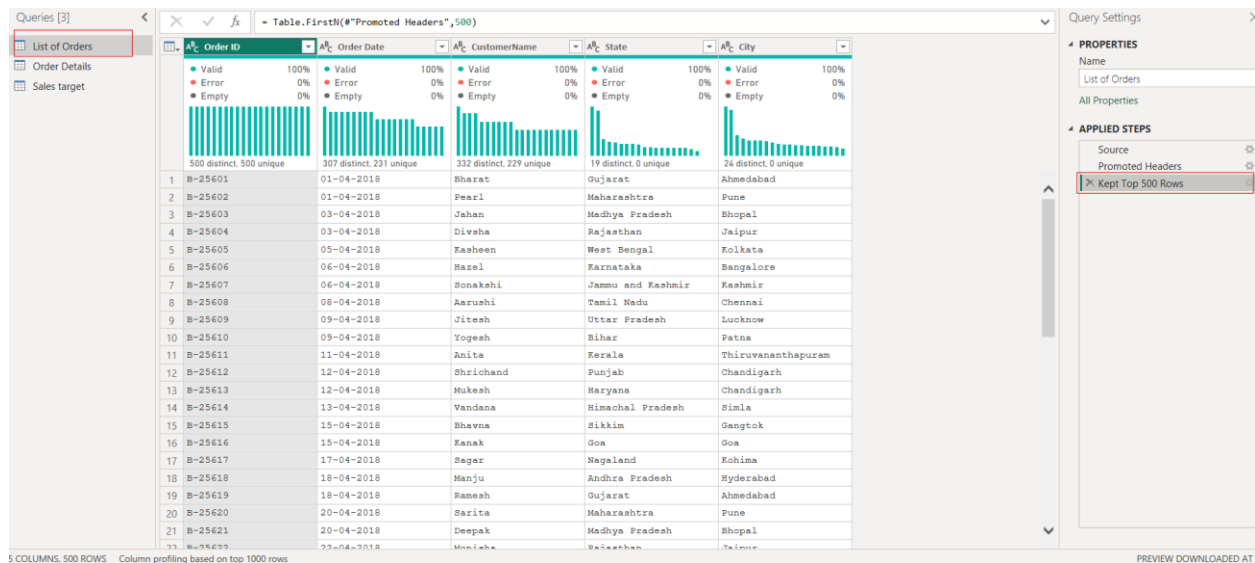
Limit to 500 rows (List of Orders only):

- Select the List of Orders query → Home → Keep Rows → Keep Top Rows → type 500 → OK.

The screenshot displays the Power Query Editor interface. The main area shows a table with columns: Order ID, Order Date, CustomerName, State, and City. The table is filtered to show the first 500 rows. A dialog box titled 'Keep Top Rows' is open, prompting the user to specify the number of rows to keep. The input field is set to 500. The dialog box also includes 'OK' and 'Cancel' buttons.

Table Data (First 7 rows):

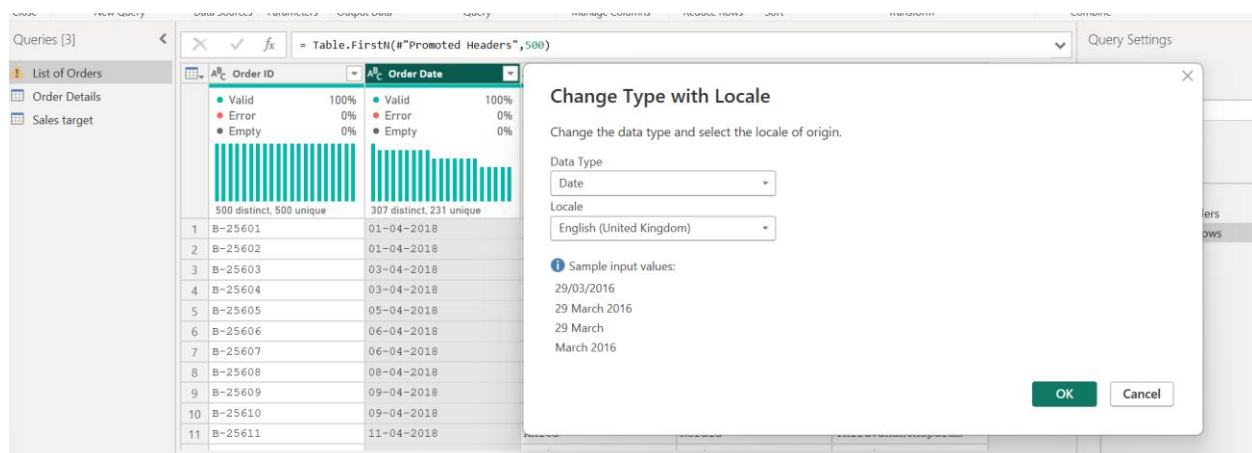
Order ID	Order Date	CustomerName	State	City
1 B-25601	01-04-2018	Bharet	Gujarat	Ahmedabad
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune
3 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4 B-25604	03-04-2018	Divaha	Rajasthan	Jaipur
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir



○ Set the Order Date column to the Date data type.

Order Date → Date type:

- Your dates are formatted dd-mm-yyyy (e.g., 01-04-2018). Power BI's default locale is US (month-first), so it may misread days above 12 as errors.
- Right-click the Order Date column header → Change Type → Using Locale → set Data Type = Date, Locale = English (United Kingdom) → OK.
- This avoids the classic mistake where 03-04-2018 gets read as March 4 instead of April 3.



Queries [3] fx = Table.TransformColumnTypes(#"Kept Top 500 Rows",{{"Order Date", type date}}, "en-GB")

Order ID	Order Date	CustomerName	State	City
1 B-25601	01/04/2018	Bharat	Gujarat	Ahmedabad
2 B-25602	01/04/2018	Pearl	Maharashtra	Pune
3 B-25603	03/04/2018	Tahan	Madhya Pradesh	Bhopal
4 B-25604	03/04/2018	Hivsha	Rajasthan	Jainpur

Query Settings

PROPERTIES

Name
List of Orders

All Properties

APPLIED STEPS

Source
Promoted Headers
Kept Top 500 Rows
Changed Order Date with Loc...
Changed Order Date with Locale

○ Change the Amount and Target columns to fixed decimal numbers.

Amount and Target → Fixed Decimal Number:

- Select the Amount column in Order Details → Transform (or right-click) → Data Type → Fixed Decimal Number.
- *Term:* Fixed Decimal Number always stores exactly 4 decimal places internally and avoids the tiny rounding errors you can get with plain "Decimal Number" (which is floating-point).

File Home Transform Add Column View Tools Help

Close New Query Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Data Type: Text

Queries [3] fx = Table.PromoteHeaders(source, [PromoteAllScalars = true])

Order ID	Amount	Profit	Quantity	Category
1 B-25601	1275.00	-1148.00	7	Furniture
2 B-25601	66.00	-12.00	5	clothing
3 B-25601	8.00	-2.00	3	clothing
4 B-25601	80.00	-56.00	4	Electronics
5 B-25602	168.00	-111.00	2	Electronics
6 B-25602	424.00	-272.00	5	Electronics

Query Settings

PROPERTIES

Name
Order Details

All Properties

APPLIED STEPS

Source
Promoted headers
Amount Changed to Fixed De...

Amount Changed to Fixed Decimal Numbers

- Do the same for Target in Sales Target.

The screenshot shows the Power Query Editor with the formula bar set to `= Table.PromoteHeaders(Source, [PromoteAllScalars = true])`. The data table has three columns: 'Month of Order Date', 'Category', and 'Target'. The 'Target' column has a value distribution of 16 distinct values. A right-click context menu is open over the 'Target' column, and the 'Change Type' option is selected, leading to a submenu where 'Text' is chosen.

Month of Order Date	Category	Target
1 Apr-18	Furniture	10400.0
2 May-18	Furniture	10500.0
3 Jun-18	Furniture	10600.0
4 Jul-18	Furniture	10800.0
5 Aug-18	Furniture	10900.0
6 Sep-18	Furniture	11000.0
7 Oct-18	Furniture	11100.0
8 Nov-18	Furniture	11300.0
9 Dec-18	Furniture	11400.0
10 Jan-19	Furniture	11500.0
11 Feb-19	Furniture	11600.0

The screenshot shows the Power Query Editor with the formula bar set to `= Table.TransformColumnTypes(#"Promoted headers",{{"Target", Currency.Type}})`. The 'Target' column is now formatted with a currency symbol. A tooltip indicates the next step: 'Target Changed to Fixed Decimal Number'.

Month of Order Date	Category	Target
1 Apr-18	Furniture	10,400.00
2 May-18	Furniture	10,500.00
3 Jun-18	Furniture	10,600.00
4 Jul-18	Furniture	10,800.00
5 Aug-18	Furniture	10,900.00
6 Sep-18	Furniture	11,000.00
7 Oct-18	Furniture	11,100.00
8 Nov-18	Furniture	11,300.00
9 Dec-18	Furniture	11,400.00

3. Text Formatting:

- Format the Customer Name column to proper case to ensure consistent capitalization.
 - Right-click Customer Name → Transform → Capitalize Each Word.

Queries [3] ✕ ✓ fx = Table.TransformColumnTypes("#Kept Top 500 Rows", {"Order Date", type date}, "en-GB")

List of Orders

Order Details

Sales target

Order ID

Order Date

Customer

City

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

500 distinct, 500 unique

307 distinct, 231 unique

332 distinct, 22

24 distinct, 0 unique

1 B-25601 01/04/2018 Bharat

2 B-25602 01/04/2018 Pearl

3 B-25603 03/04/2018 Jahan

4 B-25604 03/04/2018 Divsha

5 B-25605 05/04/2018 Kasheen

6 B-25606 06/04/2018 Hazel

7 B-25607 06/04/2018 Sonakshi

8 B-25608 08/04/2018 Aarushi

9 B-25609 09/04/2018 Jitesh

10 B-25610 09/04/2018 Yogesh

11 B-25611 11/04/2018 Anita

12 B-25612 12/04/2018 Shrichand

13 B-25613 12/04/2018 Mukesh

14 B-25614 13/04/2018 Vandana

15 B-25615 15/04/2018 Bhavna

16 B-25616 15/04/2018 Kanak

Copy

Remove

Remove Other Columns

Duplicate Column

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Split Column

Group By...

Fill

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

Drill Down

Add as New Query

lowercase

UPPERCASE

Capitalize Each Word

Trim

Clean

Length

JSON

XML

Ahmedabad

lowercase

UPPERCASE

Capitalize Each Word

Trim

Clean

Length

JSON

XML

Patna

Thiruvananthapuram

Chandigarh

Chandigarh

Simla

Gangtok

Goa

Queries [3] ✕ ✓ fx = Table.TransformColumns("#Changed Order Date with Locale", {"CustomerName", Text.Proper, type text})

List of Orders

Order Details

Sales target

Order ID

Order Date

CustomerName

State

City

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

Valid 100% Error 0% Empty 0%

500 distinct, 500 unique

307 distinct, 231 unique

332 distinct, 229 unique

19 distinct, 0 unique

24 distinct, 0 unique

1 B-25601 01/04/2018 Bharat

2 B-25602 01/04/2018 Pearl

3 B-25603 03/04/2018 Jahan

4 B-25604 03/04/2018 Divsha

5 B-25605 05/04/2018 Kasheen

6 B-25606 06/04/2018 Hazel

7 B-25607 06/04/2018 Sonakshi

8 B-25608 08/04/2018 Aarushi

9 B-25609 09/04/2018 Jitesh

10 B-25610 09/04/2018 Yogesh

11 B-25611 11/04/2018 Anita

12 B-25612 12/04/2018 Shrichand

Pujarat

Maharashtra

Madhya Pradesh

Rajasthan

West Bengal

Karnataka

Jammu and Kashmir

Tamil Nadu

Uttar Pradesh

Bihar

Kerala

Punjab

Ahmedabad

Pune

Bhopal

Jaipur

Kolkata

Bangalore

Kashmir

Chennai

Lucknow

Patna

Thiruvananthapuram

Chandigarh

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Kept Top 500 Rows

Changed Order Date with Loc...

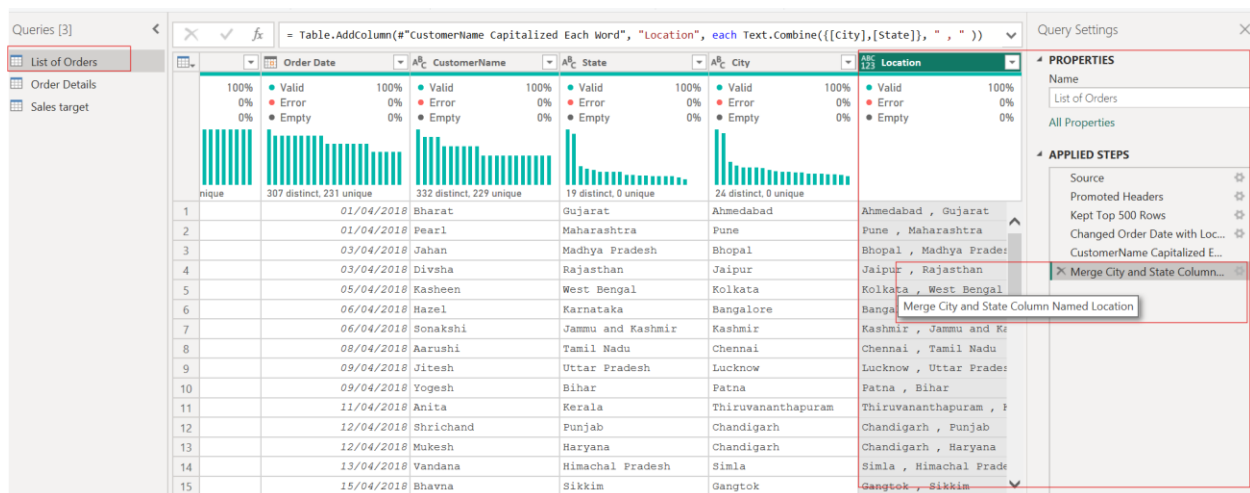
CustomerName Capitalized Each Word

4. Column Creation:

○ Merge the City and State columns to create a new column named Location in the format: City, State

Location column (City, State format):

- Add Column → Custom Column → Name: Location → Formula: Text.Combine({[City], [State]}, ", ") (Using Text. Combine avoids errors if either field is ever blank.)



- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

Profit Margin column (Profit ÷ Amount, as %):

- Add Column → Custom Column → Name: Profit Margin → Formula: [Profit] / [Amount]
- Then select that column → Transform → Data Type → Percentage, so it displays as e.g. -90.04% instead of -0.9004.



Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula ⓘ
= [[Profit]] / [Amount]

Available columns
Order ID
Amount
Profit
Quantity
Category
Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Profit, Quantity, Category, Sub-Category, and % Profit Margin. The formula bar at the top shows the custom column formula: `= Table.TransformColumnTypes(#"Added Profit Margin",{"Profit Margin", Percentage.Type})`. The right sidebar shows the 'Query Settings' for 'Order Details', with the 'APPLIED STEPS' list containing: Source, Promoted headers, Amount Changed to Fixed Dec..., Profit Changed to Fixed Dec..., Added Profit Margin, and Changed Profit Margin Type t... (highlighted with a red box). The table data includes rows with profit values, quantities, and categories like Furniture, Clothing, Electronics, and Phones.

	Profit	Quantity	Category	Sub-Category	% Profit Margin
1	1,275.00	-1,148.00	7 Furniture	Bookcases	-90.0
2	66.00	-12.00	5 Clothing	Stole	-18.1
3	8.00	-2.00	3 Clothing	Hankchief	-25.0
4	80.00	-56.00	4 Electronics	Electronic Games	-70.0
5	168.00	-111.00	2 Electronics	Phones	-66.0
6	424.00	-272.00	5 Electronics	Phones	-64.1
7	2,617.00	1,151.00	4 Electronics	Phones	43.9
8	561.00	212.00	3 Clothing	Saree	37.7
9	119.00	-5.00	8 Clothing	Saree	-4.2
10	1,355.00	-60.00	5 Clothing	Trousers	-4.4
11	24.00	-30.00	1 Furniture	Chairs	-125.0
12	193.00	-166.00	3 Clothing	Saree	-86.0
13	180.00	5.00	3 Clothing	Trousers	2.7
14	116.00	16.00	4 Clothing	Stole	13.7
15	107.00	36.00	6 Clothing	Stole	33.6

5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

- Profit = 0 → Break-Even
- Profit > 0 → Profit

- Add Column → Conditional Column → Name: Profit Status
 - If Profit is less than 0 → Output Loss
 - Else if Profit equals 0 → Output Break-Even
 - Else → Output Profit

The generated M code looks like: if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else "Profit"

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Profit Status

	Column Name	Operator	Value	Then	Output
If	Profit	is less than	0	Then	Loss
Else If	Profit	equals	0	Then	Break-Even

Add Clause

Else

Profit

OK

Cancel

Queries [3]

- List of Orders
- Order Details
- Sales target

	Quantity	Category	Sub-Category	% Profit Margin	Profit Status
1	-1,148.00	Furniture	Bookcases	-90.04%	Loss
2	-12.00	Clothing	Stole	-18.18%	Loss
3	-2.00	Clothing	Hankerchief	-25.00%	Loss
4	-56.00	Electronics	Electronic Games	-70.00%	Loss
5	-111.00	Electronics	Phones	-66.07%	Loss
6	-272.00	Electronics	Phones	-64.15%	Loss
7	1,151.00	Electronics	Phones	43.98%	Profit
8	212.00	Clothing	Saree	37.79%	Profit
9	-5.00	Clothing	Saree	-4.20%	Loss
10	-60.00	Clothing	Trousers	-4.43%	Loss
11	-30.00	Furniture	Chairs	-125.00%	Loss
12	-166.00	Clothing	Saree	-86.01%	Loss
13	5.00	Clothing	Trousers	2.78%	Profit
14	16.00	Clothing	Stole	13.79%	Profit
15	36.00	Clothing	Stole	33.64%	Profit

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted headers

Amount Changed to Fixed De...

Profit Changed to Fixed Deci...

Added Profit Margin

Changed Profit Margin Type t...

Added Conditional Column N...

Added Conditional Column Named Profit Status

6. Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.
- Home → Merge Queries (not "Merge Queries as New" — or use "as New" if you want to keep the two originals untouched, which you'll need for Task 10).
- First table: List of Orders. Second table: Order Details. Matching column: Order ID on both sides.
- Join kind: Left Outer (keeps every order even if some order details are missing) — or Inner if you only want orders that have matching line items.
- Click OK, then click the expand icon on the new merged column to bring in the Order Details fields (Sub-Category, Quantity, etc. — skip duplicating Category/Amount/Profit if you don't need both copies).
- Rename this query to Orders Data.

The screenshot displays the Power BI Desktop interface. On the left, the 'Queries' pane shows four queries: 'List of Orders', 'Order Details', 'Sales target', and 'Orders Data' (highlighted with a red box). The main workspace shows a data table with columns: 'Order ID', 'CustomerName', 'State', 'City', 'Location', and 'Order Details'. Each column has a data type icon and a status bar indicating 'Valid' (100%), 'Error' (0%), and 'Empty' (0%). The 'Order Details' column is expanded, showing a list of order details for each order ID. The 'Query Settings' pane on the right shows the 'Name' field set to 'Orders Data' and the 'Applied Steps' list containing 'Source' and 'Expanded Order Details'.

Order ID	CustomerName	State	City	Location	Order Details
1	4/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad , Gujarat
2	4/2018	Pearl	Maharashtra	Pune	Pune , Maharashtra
3	4/2018	Jahan	Madhya Pradesh	Bhopal	Bhopal , Madhya Pradesh
4	4/2018	Divsha	Rajasthan	Jaipur	Jaipur , Rajasthan
5	4/2018	Kasheen	West Bengal	Kolkata	Kolkata , West Bengal
6	4/2018	Hazel	Karnataka	Bangalore	Bangalore , Karnataka
7	4/2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir , Jammu and Ka...
8	4/2018	Aarushi	Tamil Nadu	Chennai	Chennai , Tamil Nadu
9	4/2018	Ujtesh	Uttar Pradesh	Lucknow	Lucknow , Uttar Pradesh
10	4/2018	Yogesh	Bihar	Fatna	Fatna , Bihar
11	4/2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram , K...
12	4/2018	Shrichand	Punjab	Chandigarh	Chandigarh , Punjab
13	4/2018	Mukesh	Haryana	Chandigarh	Chandigarh , Haryana
14	4/2018	Vandana	Himachal Pradesh	Simla	Simla , Himachal Prade...
15	4/2018	Bhavna	Sikkim	Gangtok	Gangtok , Sikkim

Expand Order Details

Select the columns to expand.

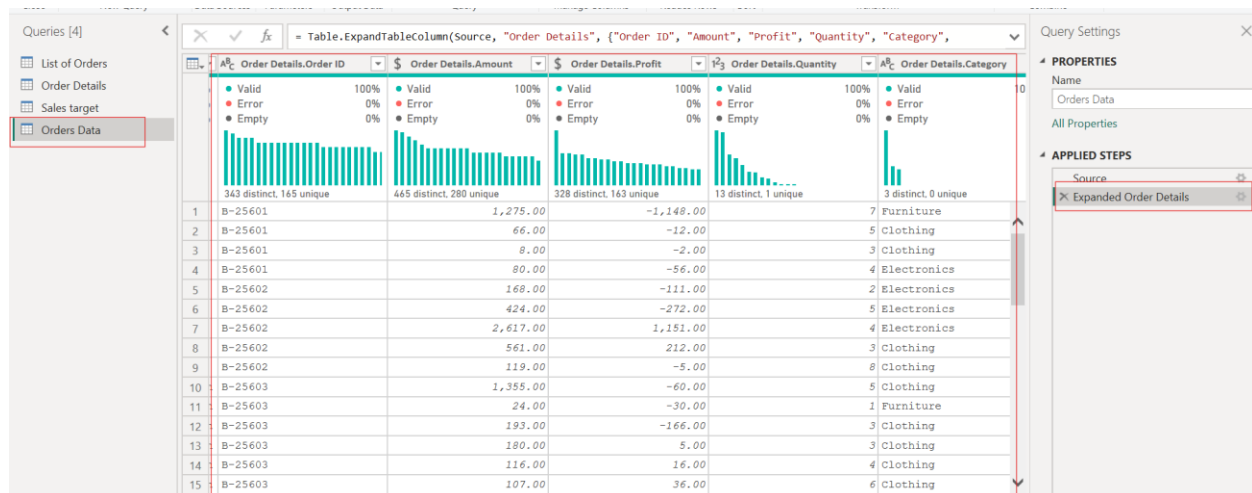


- ☒ (Select All Columns)
- ☒ Order ID
- ☒ Amount
- ☒ Profit
- ☒ Quantity
- ☒ Category
- ☒ Sub-Category
- ☒ Profit Margin
- ☒ Profit Status

Default column name prefix (optional)

OK

Cancel



7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
 - Replacing missing values
 - Removing records
 - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

Find missing values:

- View → Column Profile and turn on Column Distribution / Column Quality — this shows % empty/error/valid per column at a glance.

File	Home	Transform	Add Column	View	Tools	Help
Query Settings	☒ Formula Bar ☒ Monospaced ☒ Column distribution ☒ Show whitespace ☒ Column profile ☒ Column quality	☐ Always allow ☐ Go to Column ☐ Advanced Editor ☐ Query Dependencies				
Layout	Data Preview	Columns	Parameters	Advanced	Dependencies	

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

= Table.AddColumn(#"CustomerName Capitalized Each Word", "Location", each Text.Combine({[City],[State]}, " , "))

Order ID	Order Date	CustomerName	State	City	Location
1 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad , Gujarat
2 B-25602	4/1/2018	Pearl	Maharashtra	Pune	Pune , Maharashtra
3 B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal	Bhopal , Madhya Pradesh
4 B-25604	4/3/2018	Divsha	Rajasthan	Jaipur	Jaipur , Rajasthan
5 B-25605	4/5/2018	Kasheen	West Bengal	Kolkata	Kolkata , West Bengal
6 B-25606	4/6/2018	Hazel	Karnataka	Bangalore	Bangalore , Karnataka
7 B-25607	4/6/2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir , Jammu and Ka...
8 B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai	Chennai , Tamil Nadu
9 B-25609	4/9/2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow , Uttar Pradesh
10 B-25610	4/9/2018	Yogesh	Bihar	Patna	Patna , Bihar
11 B-25611	4/11/2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram , K...
12 B-25612	4/12/2018	Shrichand	Punjab	Chandigarh	Chandigarh , Punjab

File	Home	Transform	Add Column	View	Tools	Help
Query Settings	☒ Formula Bar ☒ Monospaced ☒ Column distribution ☒ Show whitespace ☒ Column profile ☒ Column quality	☐ Always allow ☐ Go to Column ☐ Advanced Editor ☐ Query Dependencies				
Layout	Data Preview	Columns	Parameters	Advanced	Dependencies	

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

= Table.AddColumn(#"Changed Profit Margin Type to Percentage", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else "Profit")

Order ID	Amount	Profits	Quantity	Category	Sub-Category	% Profit Margin	Profit Status
1 B-25601	1,275.00	-1,148.00	7	Furniture	Bookcases	-90.04%	Loss
2 B-25601	66.00	-12.00	5	Clothing	Stole	-18.18%	Loss
3 B-25601	8.00	-2.00	3	Clothing	Hankerchief	-25.00%	Loss
4 B-25601	80.00	-56.00	4	Electronics	Electronic Games	-70.00%	Loss
5 B-25602	168.00	-111.00	2	Electronics	Phones	-66.07%	Loss
6 B-25602	424.00	-272.00	5	Electronics	Phones	-64.15%	Loss
7 B-25602	2,617.00	1,151.00	4	Electronics	Phones	43.98%	Profit
8 B-25602	561.00	212.00	3	Clothing	Saree	37.79%	Profit
9 B-25602	119.00	-5.00	8	Clothing	Saree	-4.20%	Loss
10 B-25603	1,355.00	-60.00	5	Clothing	Trousers	-4.43%	Loss
11 B-25603	24.00	-30.00	1	Furniture	Chairs	-125.00%	Loss
12 B-25603	193.00	-166.00	3	Clothing	Saree	-86.01%	Loss
13 B-25603	180.00	5.00	3	Clothing	Trousers	2.78%	Profit
14 B-25603	116.00	16.00	4	Clothing	Stole	13.79%	Profit
15 B-25603	107.00	36.00	6	Clothing	Stole	33.64%	Profit
16 B-25603	12.00	1.00	2	Clothing	Hankerchief	8.33%	Profit
17 B-25603	38.00	18.00	1	Clothing	Kurti	47.37%	Profit

Duplicates:

- Select all columns → Home → Remove Rows → Remove Duplicates. Only do this on genuinely identical rows across every column — don't apply it to Order ID alone in Order Details, since one Order ID legitimately repeats across multiple product lines.

The image displays three screenshots of the Power BI Desktop interface, illustrating the 'Remove Duplicates' step in the 'Applied Steps' pane for different queries.

Top Screenshot: List of Orders

- Query: `Table.Distinct(#"Merge City and State Column Named Location", {"Order Date", "CustomerName", "State", "City", "Location" })`
- Columns: Order ID, Order Date, CustomerName, State, City, Location
- Applied Steps: Source, Promoted Headers, Kept Top 500 Rows, Changed Order Date with Loc..., CustomerName Capitalized E..., Merge City and State Column..., **Removed Duplicates**

Middle Screenshot: Order Details

- Query: `Table.Distinct(#"Added Conditional Column Named Profit Status", {"Order ID", "Amount", "Profit", "Category", "Sub-Category" })`
- Columns: Order ID, Amount, Profit, Category, Sub-Category
- Applied Steps: Source, Promoted headers, Amount Changed to Fixed De..., Profit Changed to Fixed Deci..., Added Profit Margin, Changed Profit Margin Type L..., **Added Conditional Column N...**, **Removed Duplicates**

Bottom Screenshot: Sales target

- Query: `Table.Distinct(#"Target changed to Fixed Decimal")`
- Columns: Month of Order Date, Category, Target
- Applied Steps: Source, Promoted headers, **Target Changed to Fixed Deci...**, **Removed Duplicates**

8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.

Sort Orders Data by Order Date descending:

- Click the dropdown arrow on the Order Date column header → Sort Descending.

The screenshot displays the Power BI Desktop interface with the 'Orders Data' table selected in the 'Queries' pane. The table is expanded, showing columns: Order ID, Order Date, CustomerName, State, City, and Location. The 'Order Date' column header has a dropdown arrow, and the 'Sort Descending' option is selected. The table data is sorted by Order Date in descending order, with the latest date (3/31/2019) at the top. The 'Properties' pane on the right shows the 'Expanded Order Details' step. The 'Applied Steps' pane on the right shows the 'Order Date Sort Descending' step.

Table Structure and Data:

Order ID	Order Date	CustomerName	State	City	Location
B-26100	3/31/2019	Hitika	Madhya Pradesh	Indore	Indore
B-26100	3/31/2019	Hitika	Madhya Pradesh	Indore	Indore
B-26099	3/30/2019	Bhishm	Maharashtra	Mumbai	Mumbai
B-26099	3/30/2019	Bhishm	Maharashtra	Mumbai	Mumbai
B-26099	3/30/2019	Bhishm	Maharashtra	Mumbai	Mumbai
B-26098	3/29/2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir
B-26098	3/29/2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir

- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Filter for a specific state (e.g., Tamil Nadu):

- Click the dropdown on the State column → uncheck Select All, check only Tamil Nadu → OK.
- (Consider doing this as a *duplicate* query, e.g. "Orders Data - Tamil Nadu," so your main model isn't permanently filtered.)

The image displays two screenshots of the Power BI Desktop interface, illustrating the steps to filter data by state.

Top Screenshot: The 'Queries' pane on the left shows the 'Orders Data' table. The main view displays a table with columns: Date, CustomerName, State, City, Location, and Order Details. A filter is applied to the 'State' column, showing a dropdown menu with 'Tamil Nadu' selected. The 'Query Settings' pane on the right shows the 'Properties' tab with 'Name' set to 'Orders Data' and 'All Properties' expanded.

Bottom Screenshot: The 'Queries' pane on the left shows the 'Orders Data' table. The main view displays a table with columns: Date, CustomerName, State, City, Location, and Order Details. The 'State' column is filtered to 'Tamil Nadu'. The 'Query Settings' pane on the right shows the 'Properties' tab with 'Name' set to 'Orders Data' and 'All Properties' expanded. The 'Applied Steps' pane shows the filter 'State Filtered to Tamil Nadu' applied to the 'Orders Data' table.

9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:

Duplicate Order Details three times (right-click the query → Duplicate) and apply one aggregation to each copy:

- Count of Order ID

- Order Count query: Home → Group By → Group by Order ID → New column name: Count of Order ID, Operation: Count Rows.

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Order ID

New column name

Count of Order ID

Operation

Count Rows

Column

OK

Cancel

The screenshot displays the Power BI Desktop interface. On the left, the 'Queries' pane shows a list of queries: 'List of Orders', 'Order Details', 'Sales target', 'Orders Data', 'Count of Order ID' (highlighted with a red box), 'Order Details (3)', and 'Order Details (4)'. The main view shows the 'Count of Order ID' query results. The table has two columns: 'Order ID' and 'Count of Order ID'. The 'Order ID' column contains values like 'B-25601', 'B-25602', etc. The 'Count of Order ID' column contains values like 4, 5, 8, etc. Above the table, there are two histograms: one for 'Order ID' showing 495 distinct values and one for 'Count of Order ID' showing 12 distinct values. The 'Query Settings' pane on the right shows the 'Properties' tab with the name 'Count of Order ID'. The 'Applied Steps' list includes 'Source', 'Promoted headers', 'Amount Changed to Fixed De...', 'Profit Changed to Fixed Deci...', 'Added Profit Margin', 'Changed Profit Margin Type t...', 'Added Conditional Column N...', 'Removed Duplicates', and 'Grouped Order Count ID' (highlighted with a red box).

○ Average Profit by Category

- Avg Profit by Category query: Group by Category → New column name: Average Profit, Operation: Average of Profit.

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name

Average Profit

Operation

Average

Column

Profit

OK

Cancel

The screenshot shows the Power BI interface with the 'Average Profit' query selected in the 'Queries' list on the left. The main view displays a table with two columns: 'Category' and 'Average Profit'. The table contains three rows of data: Furniture, Clothing, and Electronics. The 'Average Profit' column shows the average profit for each category: 8.633333333 for Furniture, 11.73801561 for Clothing, and 33.5 for Electronics. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' for the query, including 'Source', 'Promoted headers', 'Amount Changed to Fixed De...', 'Profit Changed to Fixed Deci...', 'Added Profit Margin', 'Changed Profit Margin Type t...', 'Added Conditional Column N...', 'Removed Duplicates', and 'Grouped Category to AVG Pr...'. The 'Grouped Category to AVG Pr...' step is highlighted with a red box.

Category	Average Profit
1 Furniture	8.633333333
2 Clothing	11.73801561
3 Electronics	33.5

○ Total Amount by Sub-Category

- Total Amount by Sub-Category query: Group by Sub-Category → New column name: Total Amount, Operation: Sum of Amount.

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Sub-Category

New column name

Total Amount

Operation

Sum

Column

Amount

OK

Cancel

The screenshot shows the Power BI Desktop interface. On the left, the 'Queries' pane lists several queries, with 'Total Amount by Sub-Category' selected. The main area displays a table with two columns: 'Sub-Category' and 'Total Amount'. The table contains 17 rows of data, including items like Bookcases, Stole, Hankerchief, Electronic Games, Phones, Saree, Trousers, Chairs, Kurti, T-shirt, Shirt, Leggings, Tables, Printers, Accessories, Furnishings, and Skirt. The 'Total Amount' column shows the sum of amounts for each sub-category. On the right, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' for the selected query. The steps include 'Source', 'Promoted headers', 'Amount Changed to Fixed De...', 'Profit Changed to Fixed Deci...', 'Added Profit Margin', 'Changed Profit Margin Type t...', 'Added Conditional Column N...', 'Removed Duplicates', and 'Grouped Sub-Category nam...'. The 'Grouped Sub-Category named Total Amount' step is highlighted with a red box.

• Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Duplicate Sales Target once:

- Group by Month of Order Date → New column name: Total Target, Operation: Sum of Target.

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Month of Order Date

New column name

Total Target

Operation

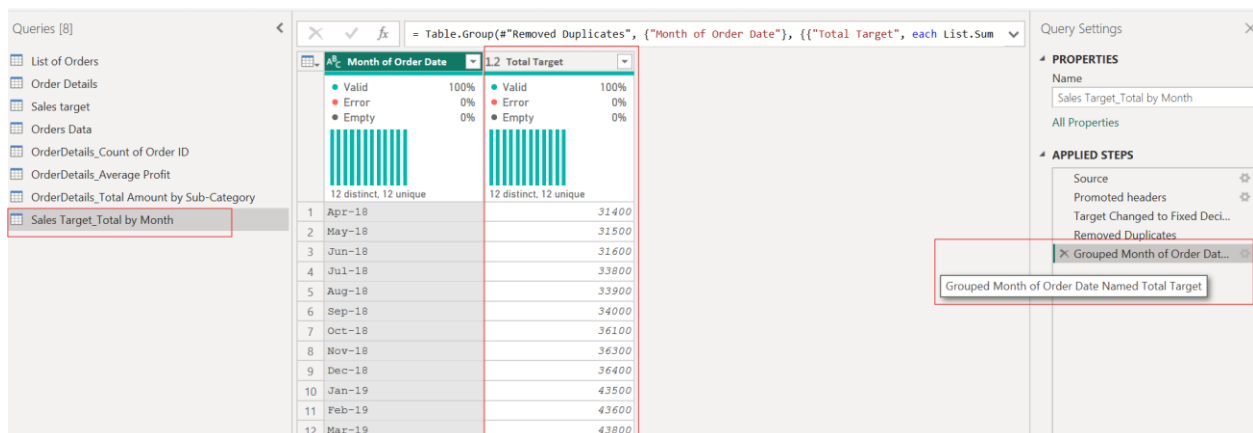
Sum

Column

Target

OK

Cancel

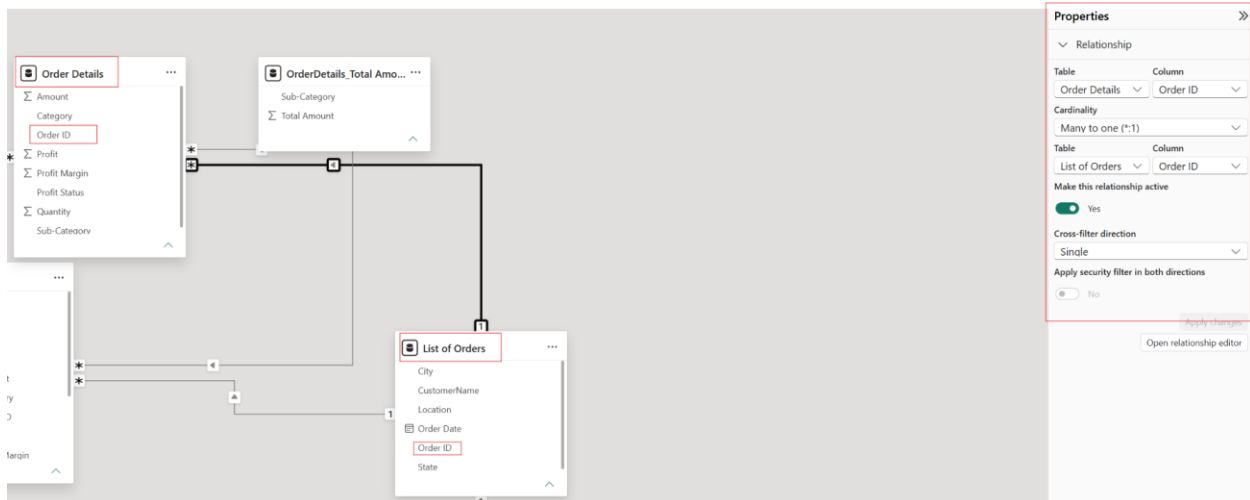


10. Data Modeling:

- Establish a relationship between:

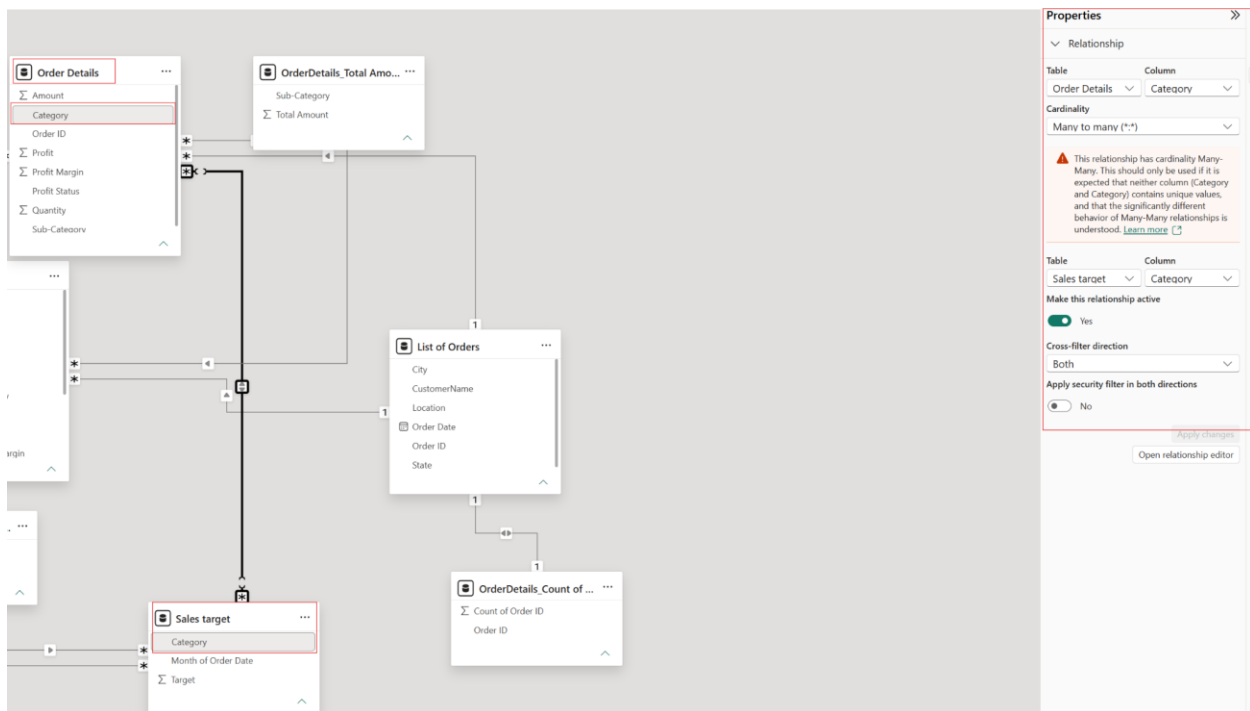
- List of Orders and Order Details using Order ID

- Click Close & Apply to load everything into the report.
- Switch to Model view (left sidebar icon that looks like connected boxes).
- Drag Order ID from List of Orders onto Order ID in Order Details → Power BI creates a relationship. Since Order ID is unique in List of Orders but repeats in Order Details, this should default to a 1-to-many relationship (1 side = List of Orders).



- Establish a relationship between:
 - Order Details and Sales Target using Category

- Drag Category from Order Details onto Category in Sales Target → another relationship (many-to-one, since Category repeats in both, verify cardinality — it may show as many-to-many; if so, check whether Category values match exactly, e.g., no trailing spaces or case differences).



2. Use Manage Relationships to ensure relationships are active and correctly defined.

- Go to Modeling → Manage Relationships → confirm both relationships appear, are marked Active (checkbox ticked), and have the correct cardinality and cross-filter direction (usually "Single" is fine for this model).

Manage relationships



+ New relationship		⚡ Autodetect	✎ Edit		🗑 Delete	≡ Filter	▼
☐	From: table (column) ↑	Relationship	To: table (column)		Status		
☐	Order Details (Category)	* ← 1	OrderDetails_Average Profit (C...		Active	...	
☐	Order Details (Category)	* ↔ *	Sales target (Category)		Active	...	
☐	Order Details (Order ID)	* ← 1	List of Orders (Order ID)		Active	...	
☐	Order Details (Sub-Category)	* ← 1	OrderDetails_Total Amount by ...		Active	...	
☐	OrderDetails_Count of Order I...	1 ↔ 1	List of Orders (Order ID)		Active	...	
☐	Orders Data (Order Details.Cat...	* ← 1	OrderDetails_Average Profit (C...		Active	...	
☐	Orders Data (Order Details.Su...	* ← 1	OrderDetails_Total Amount by ...		Active	...	
☐	Orders Data (Order ID)	* ← 1	List of Orders (Order ID)		Active	...	
☐	Sales target (Category)	* ← 1	OrderDetails_Average Profit (C...		Active	...	
☐	Sales target (Month of Order ...	* ← 1	Sales Target_Total by Month (...)		Active	...	

Close